

Online Resource 1

Supplementary material for “Quantum Cognition for Robot Decision-Making: A Systematic Review of Quantum-Like Models for Autonomous Agents”

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This resource contains the data-extraction table of the 67 application studies (Table S1), the PRISMA 2020 checklist (Table S2), the flow of records (Table S3 and Figure S1), the quality-appraisal criteria and judgements (Tables S4 and S5), the search strategy and the record of the targeted search (Table S6), the studies excluded under criterion E5 (Table S7), a glossary of symbols (Table S8) and the classification of the 71 theory and evidence studies (Table S9). Section numbers refer to the main article. Tables S1 and S5 are also provided as CSV files (Online Resources 3 and 4). Evidence levels: E1 conceptual; E2 formal model; E3 human data; E4 simulation or benchmark; E5 physical system; E6 field deployment.

Table S1: Data-extraction table of the application studies ($n = 67$): domain, model type, evidence level, task, key result, main limitation and basis of extraction (full text or abstract).
[†] also discussed in the companion survey.

Study	Domain	Model type	Level	Task or phenomenon	Key result	Main limitation	Basis
Moreira and Wichert (2014)	Decision models and networks	Quantum-like	E2	Probabilistic inference in Bayesian networks under uncertainty	Large departures from classical inference under high uncertainty; collapse to classical under low uncertainty	No human data; free interference phases	Abstract
Moreira and Wichert (2016)	Decision models and networks	Quantum-like	E3	Sure-thing principle violations (disjunction effect)	Approximates observed violations with automatically set quantum parameters	Heuristic lacks strong psychological justification; aggregate data from few paradigms	Abstract
Moreira (2017)	Decision models and networks	Quantum-like	E3	Modelling paradoxical human decisions with graphical models	Consolidates quantum-like graphical models reproducing sure-thing violations	Few paradigms; aggregate data; heuristic parameter setting	Abstract
Moreira et al. (2020b)	Decision models and networks	Quantum-like	E3	Decision-making with utilities; paradoxical choices	Predicts observed choice distributions; identifies perceived-optimal but irrational decisions; tracks belief evolution	Interference parameters still required; classic laboratory paradigms only	Abstract
Wichert et al. (2020)	Decision models and networks	Quantum-like	E3	Predicting sure-thing principle violations	Reported accurate prediction of paradoxical decisions without fitted phases	Two paradigms; aggregate data; maximum-entropy assumption	Abstract
She et al. (2021)	Decision models and networks	Quantum-like	E4	Multi-attribute decision making with interacting attributes	Reported better predictive capability than classical Bayesian network approaches	Single application; cognitive interpretation of interference unclear	Abstract
Meghdadi et al. (2022)	Decision models and networks	Quantum-like	E3	Biased human selection in decision games	RMSE 3.4 on Prisoner’s Dilemma, ranked first among compared models	Additional loosely constrained parameters; few aggregate datasets	Abstract
Gao et al. (2025)	Decision models and networks	Quantum-like	E3	Sure-thing principle violations; information retrieval	Lowest average and standard deviation of fitting errors among compared models	Existing aggregate datasets; added model complexity	Abstract
Widdows and Peters (2003)	ML, NLP and IR	Quantum-inspired	E4	Word-sense disambiguation / retrieval with logical connectives on word vectors	Negation removes unwanted senses (e.g. legal ‘suit’); operators equal quantum-logic connectives; no quantitative scores in this paper	Qualitative evaluation only; no human data; non-distributivity not exploited empirically	Abstract
Sordoni et al. (2013)	ML, NLP and IR	Quantum-inspired	E4	Ad hoc document retrieval with term dependencies	Reported competitive or improved retrieval over MRF-style dependency models (numbers not verified here)	Vocabulary-sized density matrices are costly; dependencies are hand-selected; no user/cognitive data	Abstract
Lorenz et al. (2023)	ML, NLP and IR	Quantum circuit (QPU)	E5	Sentence and noun-phrase binary classification (MC, RP)	Hardware test error 16.7% (MC) and 32.3% (RP), F-score 0.85/0.75; DisCoCat best on RP	Very small datasets, single hardware run, post-selection overhead, long queue times; no quantum advantage	Full text
Svozil (2025)	ML, NLP and IR	Quantum-inspired	E2	Contextual word embeddings / polysemy	Framework proposal; no quantitative results	No learning procedure, corpus or evaluation; practicality untested	Full text
Stoudenmire and Schwab (2016)	Deep learning and LLMs	Quantum-inspired	E4	Image classification (supervised learning)	Test-set classification error below 1% on MNIST	Single simple benchmark; 1-D ordering of 2-D inputs; no cognitive or agent evaluation	Abstract

Study	Domain	Model type	Level	Task or phenomenon	Key result	Main limitation	Basis
Zhang et al. (2018)	Deep learning and LLMs	Quantum-inspired	E4	Question answering / answer selection (text matching)	Reported effective against neural and quantum LM baselines (numbers not verified)	Interpretability rests on analogy; text-matching only; no cognitive data	Abstract
Li et al. (2019)	Deep learning and LLMs	Quantum-inspired	E4	Question answering (semantic matching)	Performance comparable to strong CNN and RNN baselines	No gain over baselines; interpretability not user-tested	Abstract
Li et al. (2021a)	Deep learning and LLMs	Quantum-inspired	E4	Multimodal video sentiment analysis (fusion)	Reported competitive with state-of-the-art fusion plus interpretability (numbers not verified)	Task-specific; interpretability argued not measured; no cognitive data	Abstract
Li et al. (2021b)	Deep learning and LLMs	Quantum-inspired	E4	Emotion recognition in conversation (multimodal)	Performance comparable to state of the art	No gain over state of the art; benchmark-only; no human-judgement validation	Abstract
Gkoumas et al. (2021)	Deep learning and LLMs	Quantum-like	E4	Multimodal decision fusion for video sentiment analysis	Substantially outperformed decision- and content-level fusion; handled cases where all unimodal classifiers failed	Incompatibility assumed not tested; benchmark labels, not human decision data	Abstract
Uprety et al. (2024)	Deep learning and LLMs	Quantum-like	E4	Order effects in similarity judgements by LLMs	LLMs show human-like order-effect bias in some settings but not others	Preliminary; prompt-dependent; no QP model fitted in abstract	Abstract
Zhang et al. (2025)	Deep learning and LLMs	Quantum-inspired	E4	Semantic matching / QA	Reported improved/competitive matching (not verified)	Details unverified; density-matrix cost; interpretability not user-tested	Abstract
Maksimovic and Maksymov (2025)	Deep learning and LLMs	Quantum-inspired	E4	Image classification with human-like confidence/uncertainty	Evidence of human-like decision-making; up to 50x faster training than classical counterpart	Human-likeness assessed via simulations, not participant data; datasets unverified	Abstract
Lo et al. (2025)	Deep learning and LLMs	Quantum-like	E4	Contextuality of LLM-derived probabilities in anaphora (co-reference) schema	77,118 (0.148%) sheaf-contextual, 36,938,948 (71.1%) Cbd-contextual; Euclidean distance best predictor; rates rise to 0.50%/81.83% for similar nouns	Single masked LM; synthetic schema; no downstream-task benefit shown	Full text
Aerts et al. (2026)	Deep learning and LLMs	Quantum-like	E4	Bell-inequality and quantum-statistics tests on LLM concept combination and text	Significant Bell-inequality violation; Bose–Einstein statistics in word distributions, mirroring human data	Prompt/version dependence; signalling not controlled in abstract; interpretive leap to convergent cognition	Abstract
Imannezhad et al. (2026)	Deep learning and LLMs	Quantum-like	E4	Probability judgement fallacies (conjunction, disjunction, complementarity) in GPT-5 vs humans	GPT-5 conjunction fallacies ~18.8% vs 58.5% humans; near-perfect complementarity; pattern closer to early QP models than noisy variants	Single proprietary LLM; personas as proxies; numbers to verify	Abstract
Kang (2026)	Deep learning and LLMs	Quantum-like	E4	Auditing question-order effects in LLM sequential judgements	Near-deterministic saturation in 17/18 and 7/8 pairs; label mapping changed QQ verdicts; no residual contextuality certified	Single small model; few items; saturated measurement interface	Abstract
Dong et al. (2008)	Reinforcement learning and agent learning	Quantum-inspired	E4	Reinforcement learning in unknown environments (gridworld navigation)	QRL learned faster than TD(0) and was robust across a much wider range of learning rates (per preprint)	Only classical simulation of the quantum algorithm; small tabular tasks; claimed parallelism speed-up requires quantum hardware	Abstract

Study	Domain	Model type	Level	Task or phenomenon	Key result	Main limitation	Basis
Dong et al. (2012) [†]	Reinforcement learning and agent learning	Quantum-inspired	E5	Mobile-robot navigation and obstacle avoidance via RL	QiRL converged in about 2000 episodes across start states (TD 1000–4000); robust for learning rates 0.01–0.12 while TD failed for ≥ 0.02 ; real robot reached target room after learning	Tabular representation; single TD baseline; real-robot results qualitative	Full text
Briegel and De las Cuevas (2012)	Reinforcement learning and agent learning	Quantum-inspired	E4	Learning agent design with internal simulation (projective simulation)	Learning curves show a trade-off between forgetting rate and achievable blocking efficiency; larger reflection time speeds learning	Toy environment only; classical simulation; no benchmark comparison with standard RL; quantum version only sketched	Full text
Melnikov et al. (2018)	Reinforcement learning and agent learning	Quantum-inspired	E4	Navigation benchmarks for RL agents (grid world, mountain car)	PS performance similar to Q-learning and SARSA; parameter tuning 1–2 orders of magnitude cheaper	Small tabular tasks; classical only; no deep RL comparison	Abstract
Wei et al. (2022)	Reinforcement learning and agent learning	Quantum-inspired	E4	Experience replay sampling in deep reinforcement learning	Outperforms DRL-PER and DCRL on most games with improved training efficiency (per abstract)	Classical heuristic only; evaluation on game benchmarks; exact gains not verified here	Abstract
Lukac and Perkowski (2007)	Robots and embodied agents	Quantum circuit (design or simulation)	E2	Generating emotional variation in humanoid robot behaviour	Architecture and examples of emotion-modulated behaviour; no quantitative results reported	No evaluation or comparison with classical stochastic models; partial testing; simulated quantum layer	Abstract
Aerts et al. (2011b)	Robots and embodied agents	Quantum-like	E1	Conceptual bridge from quantum cognition to AI and robotics	Proposes a two-layer (classical logical plus quantum conceptual) account of thought and quantum-like structure as a route for AI/robotics	No formal model for robots, no implementation or evaluation; speculative claims about computational power of the mind	Full text
Hangl et al. (2016)	Robots and embodied agents	Quantum-inspired	E5	Hierarchical manipulation skill learning by autonomous play	Robot learned sensing and preparation strategies (e.g., flipping to open box); about 80 roll-outs to reach 90% success in simulation	Small hand-designed state and skill sets; classical learning; limited task range	Abstract
Yan et al. (2019)	Robots and embodied agents	Quantum circuit (design or simulation)	E2	Representing and transforming robot emotional states	Shows emotion transitions (e.g., anger to relaxation via NOT gates) as unitary operations; argues reduced resource needs	No empirical validation, implementation or classical comparison; benefits require quantum hardware	Abstract
Moreira et al. (2020a)	Robots and embodied agents	Quantum-like	E1	Position on cognitive architectures for decision-making	Proposes quantum-like architecture as a unifying alternative to resource-rational analysis	No formal model specified in detail, no data or implementation; comparative claims not demonstrated	Abstract
Lanza et al. (2020)	Robots and embodied agents	Quantum-like	E2	Robot perception of mutually exclusive percepts	Designed behaviour reproduced in simulation; real quantum hardware showed notable errors	Binary toy scenario; no classical baseline; hardware errors; no demonstrated perceptual benefit	Full text
Hangl et al. (2020)	Robots and embodied agents	Quantum-inspired	E5	Autonomous acquisition of robotic manipulation skills through play	Behaviour composition solved previously unsolvable tasks (e.g., tower disassembly); active learning improved learning speed by about 30% in simulation	Classical PS only; hand-designed skills; supervision needed; limited scalability evidence	Abstract

Study	Domain	Model type	Level	Task or phenomenon	Key result	Main limitation	Basis
Lanza et al. (2021)	Robots and embodied agents	Quantum-like	E2	Multisensory integration and belief representation in robot perception	Measurement probabilities track similarity of inputs to target states; query operators re-map basis meanings consistently	Simulation only; no temporal integration; single query per prepared state; no classical fusion baseline	Full text
Yan et al. (2021a)	Robots and embodied agents	Quantum circuit (design or simulation)	E2	Emotion generation and transition for companion robots	Circuit schemes for generating and transitioning among Plutchik emotions (details not verified)	Apparently formal design only; no verified evaluation or classical comparison	Abstract
Ho and Hoorn (2022)	Robots and embodied agents	Quantum circuit (design or simulation)	E2	Modelling robot affect and user-directed attitude in HRI	Complete circuit framework for affective appraisal, involvement–distance and use intentions	No empirical validation or classical comparison; implementation status unclear	Full text
Song et al. (2022)	Robots and embodied agents	Quantum-like	E4	Pedestrian crossing-intention estimation and trajectory prediction for autonomous vehicles	QLB reported to capture boundedly rational crossings and edge events better than classical probability and Social-LSTM	Narrow scenario; interference parameters not clearly learned; quantitative details not verified here	Abstract
Widdows et al. (2023) [†]	Robots and embodied agents	Quantum circuit (QPU)	E5	Implementing quantum cognition models (order effects, disjunction/interference effects) as quantum circuits	Prisoner's Dilemma probabilities (82%, 73%, 64%) reproduced within 0.37%; order-effect outputs within 2–3% of expected values	No new human data or fitting; tiny circuits with no quantum advantage; hardware noise affects order-effect circuits	Full text
Mannone et al. (2024)	Robots and embodied agents	Quantum circuit (design or simulation)	E2	Modelling robot 'thinking' and creative behaviour (improvised dance)	Illustrative pose sequences, including feedback-driven unpredictable (creative-like) behaviour and 'overload' responses	Toy model; hand-set states; no behavioural evaluation or classical baseline	Abstract
Daglarli (2025) [†]	Robots and embodied agents	Quantum circuit (design or simulation)	E4	Autonomous behaviour of a smart agent (exploration, survival) in a game simulation	Reported 2× faster convergence to 80% task success and ~15% higher cumulative reward than classical deep RL baseline; gains smaller in reactive, high-uncertainty tasks	Quantum circuits classically simulated; largely qualitative reporting; unclear baselines and statistics; weak justification for algorithm choice	Full text
De Carolis et al. (2025) [†]	Robots and embodied agents	Quantum circuit (design or simulation)	E1	Review and project outline for quantum-enhanced social robotics	Identifies Grover search, quantum walks, PQC and quantum-inspired emotion models as main tools; recommends hybrid classical–quantum designs given hardware limits	No implementation or evaluation; conflates quantum hardware, quantum-like and quantum-inspired work; limited critical appraisal	Full text
Chella et al. (2026)	Robots and embodied agents	Quantum circuit (design or simulation)	E4	Reactive action selection from robot sensor data (wall following)	Two-sensor configurations best; circuit accuracy about 0.91 (weighted F1 about 0.90); errors near region boundaries	Offline only; no closed-loop robot control; no quantum advantage claimed; hand-defined rule partitions	Abstract
Yukalov et al. (2018)	Multi-agent systems and teams	Quantum-like	E2	Multistep collective decision-making; dynamic disjunction effect	Predicts initial probabilistic deviation, convergence of prospect probabilities to utility factors, and consensus after repeated information exchange; reported to match empirical observations	Stylised interaction rules; validated only against previously published aggregate data	Abstract

Study	Domain	Model type	Level	Task or phenomenon	Key result	Main limitation	Basis
Ried et al. (2019)	Multi-agent systems and teams	Quantum-inspired	E4	Collective motion / swarm alignment (marching locusts)	Reproduces density-dependent regimes (disorder, switching alignment, stable alignment); responses converge in $\sim 10^2$ steps; drift/diffusion qualitatively match experiments	1-D toy world with two actions; qualitative match only; PS is classical here	Abstract
López-Incera et al. (2020)	Multi-agent systems and teams	Quantum-inspired	E4	Swarm foraging; emergence of collective motion	Distant food: strong alignment (order parameter $\approx 0.8+$), Lévy-like trajectories; near food: weak alignment (≈ 0.2), Brownian trajectories	1-D world, two actions, simple reward; PS run classically	Abstract
Lawless (2020)	Multi-agent systems and teams	Quantum-like	E1	Design and control of autonomous human-machine teams	Argues that interdependence behaves quantum-like and that machines in A-HMTs must communicate causally; no quantitative model fit	Conceptual; analogies not formalised or tested quantitatively; case-study interpretation is post hoc	Abstract
Yan et al. (2021b) [†]	Multi-agent systems and teams	Quantum circuit (design or simulation)	E2	Representing, transforming and fusing emotional states of social/multi-robot systems	Illustrative probability distributions of fused emotions before and after controlled rotations; qubit count grows logarithmically with robot number but preparation cost is $O(2^{4n})$	Conceptual; simulation-only toy examples; no hardware execution, no empirical grounding of emotion model; exponential preparation cost and repeated measurement needed for read-out	Full text
Yukalov et al. (2022)	Multi-agent systems and teams	Quantum-like	E2	Collective decision dynamics in agent networks	Short-memory agents show prolonged probability oscillations; collective information field damps oscillations and forces convergence; reproduces dynamic disjunction effect	Idealised network and communication; no empirical calibration reported in abstract	Abstract
Yukalov (2023)	Multi-agent systems and teams	Quantum-like	E2	Affective (emotion-aware) decision-making for AI	Argues affective AI based on cognition-emotion duality avoids classical behavioural paradoxes	No implementation or empirical test; architecture unspecified	Abstract
Lawless et al. (2023) [†]	Multi-agent systems and teams	Quantum-like	E1	Structure and performance of human-machine teams in open systems	Argues team performance is non-decomposable; proposes trade-off between structural and productive entropy production	Mainly review/position; analogical formalism; no new controlled data	Full text
Essalmi et al. (2025)	Multi-agent systems and teams	Quantum-inspired	E4	Interaction-aware decision-making for automated driving (merging, roundabouts)	QG-G4 reported lower collision and higher success rates than baselines; both quantum models give higher expected payoffs than classical games under some parameters	Simulation only; 2×2 games; parameter-dependent; collision results reported as not reproducible by an independent review	Abstract
Kong et al. (2026)	Multi-agent systems and teams	Quantum-inspired	E4	Group navigation and queue formation in swarms	Polarisation ≈ 0.99 within 100 steps; peak elongation ratio 8.31; optimum at moderate granularity ($m = 15$); polarisation > 0.97 under noise up to 30°	Simulation only; PS classical; no physical robots	Abstract

Study	Domain	Model type	Level	Task or phenomenon	Key result	Main limitation	Basis
Ashtiani and Abdollahi Azgomi (2019)	Human-AI trust	Quantum-like	E4	Computational trust evolution in multi-agent/decision-support systems	Lower MAE and RMSE than BLADE, TidalTrust, max-min, max-mean and Bayesian methods; malicious recommender amplitude ≈ 0.0009 after 10 iterations	Small synthetic scenarios; no human data; no quantum speed-up	Abstract
Pushparaj et al. (2021)	Human-AI trust	Quantum-inspired	E3	Human-automation trust in air traffic control	Acceptance rate under ambiguous conditions 40% per participant; model claimed more robust than questionnaire-based models	Very small sample; no reported quantitative model comparison; limited generalisability	Abstract
Humr et al. (2023)	Human-AI trust	Quantum-like	E2	Temporal evolution of human trust in AI-supported decisions	Proposes approach; no quantitative results in abstract	Short proceedings paper; validation not reported in abstract	Abstract
Roeder et al. (2023)	Human-AI trust	Quantum-like	E3	Trust/reliability calibration in human-AI interaction	Quantum model predicted evolving reliability ratings better than Markov model; clear ERP when expectations were perturbed	Exploratory; N = 7; scripted AI behaviour	Abstract
Humr et al. (2025) [†]	Human-AI trust	Quantum-like	E2	Human-in-the-loop decision-making with AI support	Illustrative simulations of oscillating confidence; cites ~15-point question-order effect captured by QP models	No new empirical data; proposals not implemented in an AI system	Full text
van der Meer et al. (2025)	Human-AI trust	Quantum-like	E3	Dynamics of human trust/reliability ratings in human-AI interaction	Interaction-dependent Hamiltonians reported as a promising means to model trust evolution; no numbers in abstract	Preprint; quantitative evidence not given in abstract; small prior dataset	Full text
Humr and Canan (2026)	Human-AI trust	Quantum-like	E1	AI-supported (military) decision-making; human-AI alignment	Three GenAI applications failed to identify conjunction fallacies; QCAL proposed with research agenda	QCAL not implemented or evaluated; GenAI probe small and qualitative	Abstract
Kak (2018)	Decision models and networks	Quantum-like	E2	Order effects in memory and belief queries of intelligent agents	Argues that inaccessible agent states may be best modelled as quantum states; proposes tests on large medical-treatment databases	No data, implementation or quantitative comparison	Abstract
Hoorn and Ho (2019)	Robots and embodied agents	Quantum circuit (design or simulation)	E2	Concurrent (mixed) emotions and reflection in robot affect	Circuit examples of robot behaviour under superposed affective states	No simulation results, robot or user study; physical claims about neural quantum dynamics are speculative	Abstract
Essalmi et al. (2026) [†]	Multi-agent systems and teams	Quantum-inspired	E4	Interaction-aware decision-making for automated driving with several players	Higher success rates and fewer collisions than classical approaches, especially in highly interactive scenarios	Simulation only; no human drivers or vehicle tests	Abstract
Nebli et al. (2026)	Deep learning and LLMs	Quantum-inspired	E2	Sequence (language) modelling with interference between token paths	Born-rule decoding that keeps phase information is provably more expressive on the constructed task family	No experiments; relevance of interference to natural language untested	Abstract

Table S2: PRISMA 2020 checklist: location of each item in the main article and this resource.

Item	Topic	Location
1	Title	Title page (identified as a systematic review)
2	Abstract	Abstract (unstructured, as required by the journal)
3	Rationale	Sections 1.1–1.4
4	Objectives	Section 2.1 (RQ1–RQ5)
5	Eligibility criteria	Section 2.3 (I1–I3, E1–E6)
6	Information sources	Section 2.2; Table S6 (sources and dates)
7	Search strategy	Section 2.2; Table S6 (concept blocks of stage 3; exact queries of stage 4)
8	Selection process	Section 2.4
9	Data collection process	Section 2.5
10a	Data items (outcomes)	Section 2.5; Table S1; Online Resource 3
10b	Data items (other variables)	Sections 2.5 and 2.6 (model type, domain, publication status, basis)
11	Study risk of bias assessment	Section 2.5 (criteria Q1–Q7); Tables S4 and S5
12	Effect measures	Not applicable: heterogeneous tasks and metrics; no pooled effect (Section 2.6)
13a–f	Synthesis methods	Narrative synthesis by domain and evidence level (Sections 2.6 and 14.1); meta-analysis, heterogeneity and sensitivity analyses not applicable
14	Reporting bias assessment	Section 2.6 (threats to validity: publication bias)
15	Certainty assessment	Section 2.6 (evidence levels E1–E6, Table 2)
16a	Study selection	Sections 2.2–2.4; Figure S1; Table S3
16b	Excluded studies	Section 2.3; Tables S6 and S7
17	Study characteristics	Table S1; Online Resource 3
18	Risk of bias in studies	Section 14.1; Table S5; Online Resource 4
19	Results of individual studies	Table S1; Sections 7–13
20a–d	Results of syntheses	Sections 5 and 7–13; Section 14.1 and Figure 8
21	Reporting biases	Section 2.6
22	Certainty of evidence	Section 14.1; Table 2
23a–d	Discussion	Sections 14–15 and 18; limitations of the review in Section 2.6
24a	Registration	Section 2 (retrospective OSF registration)
24b	Protocol	Section 2 (OSF registration)
24c	Amendments	Sections 2.2 and 2.3: criterion E5 and the stage 4 search were added after the first corpus was assembled
25	Support	Declarations (no external funding)
26	Competing interests	Declarations
27	Availability of data, code and other materials	Declarations; Online Resources 2–4

Table S3: Flow of records. Counts for stages 1–3 are works retained after verification; the number of records each search returned and the number excluded at each screening step were not logged and cannot be reported.

Stage	Count
Stage 1: curated reading list, checked against publisher, arXiv or repository records	102 items listed; about 20 corrected, 3 unlocatable items replaced by 2 verified papers, 2 non-existent titles replaced, 2 dropped
Stage 2: authors' PDFs identified by first-page text	31 works added
Stage 3: structured searches and citation chasing (two streams)	74 works added; records not logged
Corpus after stages 1–3	205 works
Stage 4: targeted search (4 October 2026) and companion-survey cross-check	17 queries; 10 new records + 1 record; 4 included, 2 excluded, 5 not retrieved (Table S6)

Works assessed against I1–I3 and E1–E6	209
Excluded as primary studies under E5 (cited once; reviewed in the companion survey)	11 (Table S7)
Application studies (I1), graded	67
of which appraised against Q1–Q7 (levels E2–E6)	61
Theory and evidence studies (I2), graded	71
Reviews and background works (I3), not graded	17 and 43
Methodological references	3
Total works cited	212

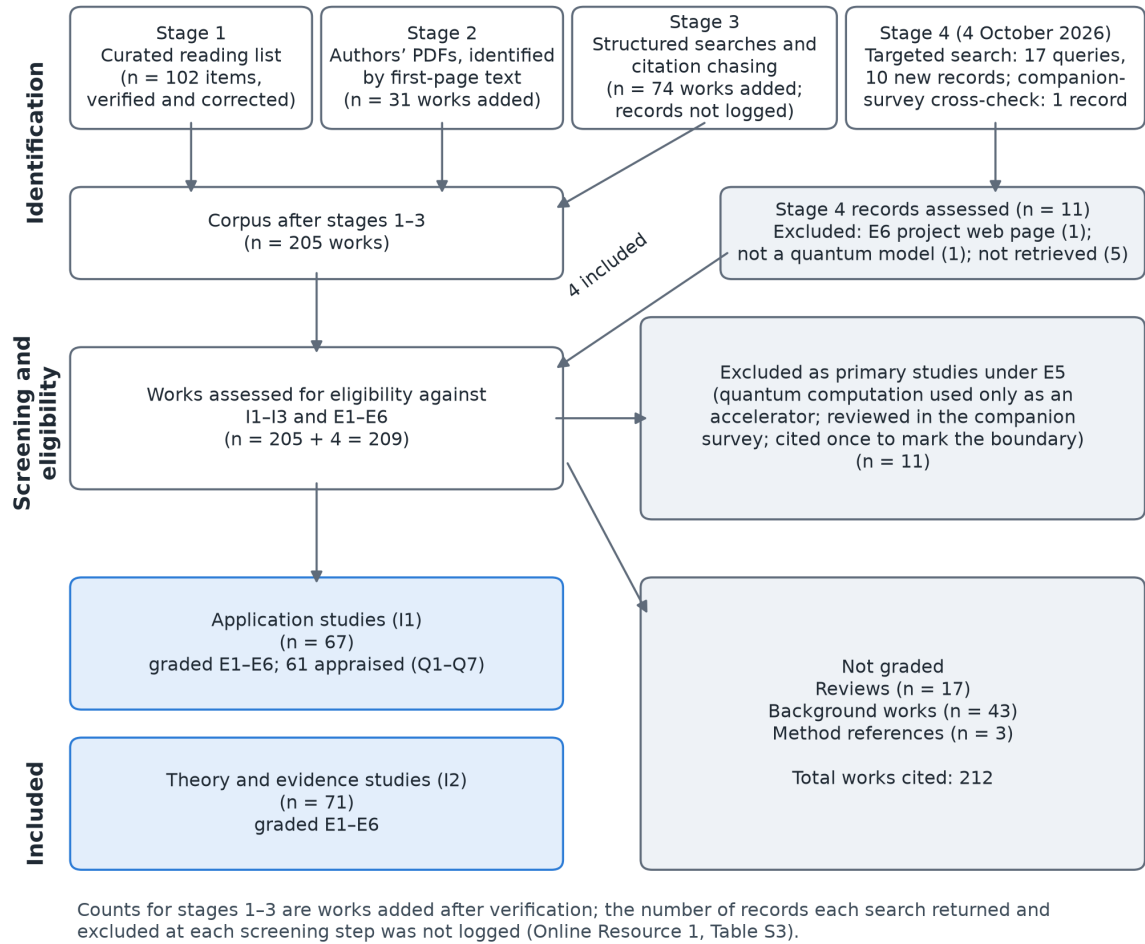


Figure S1: Flow of records, following the PRISMA 2020 flow diagram. Stages whose record counts were not logged are marked as such (Table S3).

Table S4: Quality-appraisal criteria for application studies. Each criterion was judged met (M), partly met (P) or not met (N); the judgements were not summed into a score. Q6 cannot be judged from an abstract and is marked “?” when the full text was not available.

ID	Criterion	Met (M)	Partly met (P)	Not met (N)
Q1	Model specification	State space, state preparation, operators (projectors, instruments, unitaries or Hamiltonians) and free parameters stated.	Model only sketched, or parameters not all stated.	Model not specified.

Q2	Classical comparison	A classical Bayesian, Markov, utility or neural model evaluated on the same task and data.	Classical model discussed but not evaluated on equal terms, or only weak baselines.	No classical comparison.
Q3	Empirical grounding	Human behavioural or neural data, or a realistic agent or AI task.	Narrow benchmark, aggregate data from a few classic paradigms, or a single small case.	Toy example or no data.
Q4	Parameter discipline	Parameters predicted or constrained a priori (e.g., QQ test, phase heuristics), or the model validated on held-out data.	Some parameters constrained, others fitted to the data they explain.	Free parameters fitted to the data they explain, or no evaluation.
Q5	Quantitative evaluation	Metrics reported with variability (repetitions, confidence intervals or tests).	Metrics without variability, or partly qualitative.	Qualitative only.
Q6	Reproducibility	Code and data released.	Code or data released, or the model fully specified in the paper so that it can be re-implemented.	Neither released (“available on request” counts as not met).
Q7	Critical discussion	Limitations and classical alternative explanations discussed in depth.	Limitations mentioned briefly.	No discussion of limitations.

Table S5: Quality appraisal of the 61 application studies at levels E2–E6 (the six conceptual E1 studies are not appraised). M = met; P = partly met; N = not met; ? = not assessable without the full text. Basis: source of the judgements.

Study	Basis	Q1	Q2	Q3	Q4	Q5	Q6	Q7
Moreira and Wichert (2014)	Abstract	M	M	N	N	P	?	P
Moreira and Wichert (2016)	Abstract	M	P	P	M	P	?	P
Moreira (2017)	Abstract	M	P	P	P	P	?	P
Moreira et al. (2020b)	Abstract	M	M	P	P	P	?	P
Wichert et al. (2020)	Abstract	M	P	P	M	P	?	P
She et al. (2021)	Abstract	M	M	P	P	P	?	N
Meghdadi et al. (2022)	Abstract	M	M	P	N	M	?	P
Gao et al. (2025)	Abstract	M	M	P	P	M	?	P
Widdows and Peters (2003)	Abstract	M	N	P	N	N	?	P
Sordoni et al. (2013)	Abstract	M	M	M	M	M	?	P
Lorenz et al. (2023)	Full text	M	P	P	M	M	M	M
Svozil (2025)	Full text	P	N	N	N	N	N	P
Stoudenmire and Schwab (2016)	Abstract	M	P	M	M	M	?	P
Zhang et al. (2018)	Abstract	M	M	M	M	M	?	P
Li et al. (2019)	Abstract	M	M	M	M	M	?	P
Li et al. (2021a)	Abstract	M	M	M	M	M	?	P
Li et al. (2021b)	Abstract	M	M	M	M	M	?	P
Gkoumas et al. (2021)	Abstract	M	M	M	M	M	?	P
Uprety et al. (2024)	Abstract	P	P	M	N	P	?	P
Zhang et al. (2025)	Abstract	P	M	M	M	M	?	?
Maksimovic and Maksymov (2025)	Abstract	M	M	N	N	P	?	P
Lo et al. (2025)	Full text	M	N	M	M	M	M	M
Aerts et al. (2026)	Abstract	P	N	P	M	P	?	P
Imannezhad et al. (2026)	Abstract	M	M	M	M	M	?	M
Kang (2026)	Abstract	M	M	P	M	M	?	M
Dong et al. (2008)	Abstract	M	M	N	P	P	?	P
Dong et al. (2012)	Full text	M	P	M	P	P	N	P
Briegel and De las Cuevas (2012)	Full text	M	N	N	N	P	N	P

Study	Basis	Q1	Q2	Q3	Q4	Q5	Q6	Q7
Melnikov et al. (2018)	Abstract	M	M	P	P	M	?	P
Wei et al. (2022)	Abstract	M	M	M	M	M	?	P
Lukac and Perkowski (2007)	Abstract	P	N	N	N	N	?	N
Hangl et al. (2016)	Abstract	M	P	M	P	M	?	P
Yan et al. (2019)	Abstract	M	N	N	N	N	?	N
Lanza et al. (2020)	Full text	M	N	P	N	P	P	P
Hangl et al. (2020)	Abstract	M	P	M	P	M	?	P
Lanza et al. (2021)	Full text	M	N	P	N	P	M	P
Yan et al. (2021a)	Abstract	P	N	N	N	N	?	?
Ho and Hoorn (2022)	Full text	M	P	N	N	N	N	P
Song et al. (2022)	Abstract	P	M	M	N	P	?	P
Widdows et al. (2023)	Full text	M	P	P	N	M	P	M
Mannone et al. (2024)	Abstract	P	N	N	N	N	?	P
Daglarli (2025)	Full text	P	P	P	N	P	N	N
Chella et al. (2026)	Abstract	M	M	P	M	M	?	M
Yukalov et al. (2018)	Abstract	M	P	P	P	P	?	P
Ried et al. (2019)	Abstract	M	P	P	P	P	?	P
López-Incera et al. (2020)	Abstract	M	N	P	P	M	?	P
Yan et al. (2021b)	Full text	M	N	N	N	P	N	P
Yukalov et al. (2022)	Abstract	M	N	N	P	P	?	P
Yukalov (2023)	Abstract	P	N	N	N	N	N	P
Essalimi et al. (2025)	Abstract	M	M	P	N	M	?	P
Kong et al. (2026)	Abstract	M	N	P	P	M	?	P
Ashtiani and Abdollahi Azgomi (2019)	Abstract	M	M	P	N	M	?	P
Pushparaj et al. (2021)	Abstract	P	N	M	N	P	?	P
Humr et al. (2023)	Abstract	P	N	N	N	N	?	P
Roeder et al. (2023)	Abstract	M	M	M	P	M	?	P
Humr et al. (2025)	Full text	M	P	N	N	P	N	P
van der Meer et al. (2025)	Full text	M	N	M	P	M	N	P
Kak (2018)	Abstract	P	P	N	N	N	?	P
Hoorn and Ho (2019)	Abstract	P	N	N	N	N	?	P
Essalimi et al. (2026)	Abstract	M	M	P	N	M	?	P
Nebli et al. (2026)	Abstract	M	N	N	M	N	?	M

Totals (met / partly met / not met / not assessable): Q1 47/14/0/0; Q2 24/16/21/0; Q3 19/24/18/0; Q4 18/16/27/0; Q5 27/22/12/0; Q6 3/2/9/47; Q7 7/48/4/2.

Search strategy

Stages 1–3. Sources: Google Scholar, Semantic Scholar, arXiv, PubMed and publisher sites, with backward and forward citation chasing from the main reviews (Pothos and Busemeyer, 2022; Huang et al., 2025b; Widdows et al., 2021; Meyer et al., 2022). The two search streams covered (a) applications of quantum cognition to robots, agents, human–machine teams and reinforcement learning, and (b) quantum-cognition theory, critiques, machine learning, large language models and neuroscience published from 2016 to 2026. The exact strings and record counts were not logged. To support replication, the following concept blocks reproduce the scope of the two streams; they are a reconstruction, not a log:

```
("quantum cognition" OR "quantum-like" OR "quantum probability" OR "quantum decision theory" OR
"quantum-inspired" OR "projective simulation")
AND(robot* OR agent* OR "human-robot" OR autonomous OR "multi-agent" OR swarm OR "reinforcement
learning" OR "machine learning" OR "language model*" OR trust OR automation)
or, for stream (b), AND ("order effect*" OR "conjunction fallacy" OR "disjunction effect" OR "sure-thing"
OR interference OR contextuality OR "Bayesian network*" OR critique)
```

Stage 4 (4 October 2026). Exact queries, results and decisions are given in Table S6.

Table S6: Targeted search of 4 October 2026: queries and decisions. The full log, including the verification queries, is in the method workbook (targeted_search_record.md).

#	Query	New records and decisions
T1	quantum-like model human-robot interaction decision-making robot 2024 2025	Genoa research-project page (excluded, E6)
T2	quantum cognition large language model order effects 2025 arXiv quantum probability	Kak (2018) and Nebli et al. (2026) (included); arXiv:2510.13894 (not retrieved)
T3	quantum probability model trust automation human-autonomy teaming 2024 2025 study	Flight-test trust case study (excluded, not a quantum model); arXiv:2410.20496 and arXiv:2503.16227 (not retrieved)
T4	quantum-like Bayesian network decision support autonomous driving or robot 2023 2024 2025	No new record
T5	quantum decision theory artificial agents reinforcement learning quantum-like agent Yukalov 2024 2025	No new record
T6	quantum-inspired cognitive model robot emotion social robot 2024 2025 quantum-like	Hoorn and Ho (2019) (included); J. Phys.: Conf. Ser. 2115, 012040 (not retrieved)
V1–V11	Eleven verification queries (titles, authors and abstracts of the records above)	arXiv:2608.14691 seen as a title only (not retrieved)
Other	Cross-check against the companion survey’s corpus	Essalmi et al. (2026) (included)

Table S7: Studies excluded as primary studies under criterion E5 (quantum computation used only to accelerate a robot or learning task, without a cognitive or decision model). They are reviewed in the companion survey (Guru and Chauhan, 2026) and cited once in the main article to mark the boundary.

Study	Method
Chen et al. (2020)	Variational quantum circuits as Q-function approximators
Jerbi et al. (2021)	Parametrised quantum policies; learning separations on constructed tasks
Skolik et al. (2022)	Variational deep Q-learning (CartPole)
Acuto et al. (2022)	Variational quantum soft actor-critic for a simulated arm
Sinha et al. (2025)	Quantum critic for self-driving navigation (Nav-Q)
Hohenfeld et al. (2024)	Variational deep Q-learning for robot navigation
Yun et al. (2022)	Quantum multi-agent reinforcement learning
Paparo et al. (2014)	Quantum speed-up of agent deliberation (quantum walks)
Chella et al. (2022)	Grover search for robot motion planning
Mannone et al. (2022)	Quantum-circuit models of swarm interaction rules
Khoshnoud et al. (2020)	Quantum communication for cooperating robots

Table S8: Glossary of symbols and notation.

Symbol	Meaning
$ \psi\rangle, \langle\psi $	State vector (ket) and its conjugate transpose (bra)
$\langle\phi \psi\rangle$	Inner product
ρ	Density matrix (mixed state); $\text{tr } \rho = 1, \rho \succeq 0$
P_A	Projector onto the “yes” subspace of question A
$I - P_A$	Projector onto the “no” subspace
$[A, B] = AB - BA$	Commutator; zero for compatible questions
$\ P \psi\rangle\ ^2$	Born-rule probability

$U(t) = \exp(-iHt)$	Unitary evolution generated by Hamiltonian H
L_k, γ_k	Lindblad (jump) operators and rates in open-system models
$\otimes$	Tensor product of spaces or states
θ	Interference phase between two branches
QQ	Quantum question equality
QLBN	Quantum-like Bayesian network
QRL, QiRL	Quantum and quantum-inspired reinforcement learning
PQC, VQC	Parametrised and variational quantum circuit
PS	Projective simulation
CbD	Contextuality-by-default
POVM	Positive-operator-valued measure (generalised measurement)
QPU	Quantum processing unit

Table S9: Theory and evidence studies ($n = 71$), graded on the evidence scale: phenomenon, key result and basis of extraction.

Study	Level	Phenomenon	Key result	Basis
Khrennikov (1999)	E2	Formal description of cognitive and social information processes	No empirical results; provides a p-adic formalism for information dynamics	Full text
Aerts et al. (2000)	E2	Logical structure of disjunction; concept combination	Quantum disjunction shown non-equivalent to classical 'or' owing to EPR-like correlations	Abstract
Busemeyer et al. (2006)	E2	Choice and response-time dynamics in decision-making	Quantum dynamics produce interference effects not possible under Markov models; predictions contrasted	Abstract
Pothos and Busemeyer (2009)	E3	Sure-thing principle violations in prisoner's dilemma and two-stage gambles	Interference reproduces violations of the law of total probability that classical models cannot	Abstract
Busemeyer et al. (2009)	E3	Choosing between Markov and quantum dynamic models of decision	Provides parameter-free tests discriminating Markov vs quantum models; evidence reviewed on both properties	Abstract
Aerts (2009)	E3	Concept combination; membership weights for conjunctions and disjunctions	Data shown incompatible with classical models; quantum model reproduces over- and underextension	Full text
Khrennikov (2010)	E2	Foundations of quantum-like modelling in psychology, economics and finance	Systematic quantum-like framework linking contextuality and interference to QP representations	Abstract
Trueblood and Busemeyer (2011)	E3	Order effects in sequential evidence evaluation (inference)	Quantum model reproduced the pattern of order effects better than the anchoring-and-adjustment model (per Bruza 2015)	Abstract
Busemeyer et al. (2011)	E3	Probability judgement fallacies (conjunction, disjunction, inverse, conditional)	Quantum model explains more of 15 major findings than three rival theories	Full text
Aerts and Sozzo (2011)	E3	Concept combination; entanglement between concepts	CHSH value $2.4197 > 2$, interpreted as violation of Bell's inequality	Full text
Aerts et al. (2011c)	E2	Concept combination, decision paradoxes and information retrieval	Argues quantum structure is a structural model for human and artificial thought; proposes SCoP-based knowledge organisation	Full text
Aerts et al. (2011a)	E3	Sure-thing principle violations; disjunction effect (Hawaii problem)	Quantum model reproduces 54%/57% vs 32% choice pattern as interference of conceptual landscape	Full text
Yukalov and Sornette (2011)	E3	Decision under risk and uncertainty; classical paradoxes	Explains disjunction effect, conjunction fallacy, Allais and Ellsberg-type paradoxes	Abstract
Busemeyer and Bruza (2012)	E3	Order effects, judgement errors, concepts, memory, decision dynamics	Unified QP account of multiple anomalies; comparisons with classical/Markov models	Full text
Pothos and Busemeyer (2013)	E3	General case for QP cognitive modelling	Argues many judgement and decision anomalies have natural QP accounts	Full text
Wang et al. (2013)	E1	Programmatic case for quantum models of cognition	Frames QP as a general cognitive modelling framework and lists open questions	Abstract
Wang and Busemeyer (2013)	E3	Question order effects in attitude surveys	QQ equality supported in the data examined	Abstract
Aerts et al. (2013)	E3	Concept combination and conceptual dynamics	Quantum models reproduce guppy-type effects, over/underextension and entanglement	Abstract
Wang et al. (2014)	E3	Question order effects in national surveys	QQ equality statistically supported; anchoring-adjustment models fail to reproduce patterns	Abstract
Khrennikov et al. (2014)	E2	Consistency of quantum measurement models with order effects and response replicability	Projective quantum models cannot jointly reproduce order effects and response replicability	Abstract

Study	Level	Phenomenon	Key result	Basis
Bruza et al. (2015)	E3	Review of quantum vs Bayesian models of judgement and decision	QP accounts for order effects, QQ equality (70 surveys), total-probability violations; classical and QP coincide for compatible events	Full text
Busemeyer and Wang (2015)	E1	Introduction to quantum cognition for psychologists	Summarises QP explanations of order effects, fallacies and sure-thing violations	Abstract
Kvam et al. (2015)	E3	Choice-confidence interference in perceptual evidence accumulation	Intermediate choice changed later confidence (interference); quantum walk fitted better than Markov	Abstract
Yearsley and Busemeyer (2016)	E2	Tutorial on constructing quantum cognitive models	Step-by-step construction of order-effect models and derivation of the QQ equality	Abstract
Wang and Busemeyer (2016)	E3	Interference of categorisation on decision-making	Interference (violation of total probability) in some conditions; BAE model predicted the pattern	Abstract
Yearsley and Pothos (2016)	E3	Effect of repeated judgements on belief change (quantum Zeno effect)	Frequent judgements associated with slower belief change, consistent with a Zeno effect	Abstract
Yukalov and Sornette (2016)	E2	Foundations of quantum probability for measurement and decision-making	Predicted 0.35/0.65 vs observed 0.37/0.63 cooperation/defection; conditions for classical reduction	Full text
Dzhafarov and Kujala (2016)	E2	Defining and testing contextuality in systems of random variables	Criterion distinguishing true contextuality from direct context influences on marginals	Abstract
Yearsley (2017)	E2	Tutorial on advanced quantum-cognition tools	Framework for incorporating noise and imperfect measurement in QP models	Abstract
Pothos et al. (2017)	E2	Normative (rational) status of quantum probabilistic judgement	Argues QP reasoning can be principled rather than simply erroneous under incompatible perspectives	Abstract
Trueblood et al. (2017)	E3	Probabilistic inference judgements (order effects, inverse fallacy, Markov violations, anti-discounting)	Accounts for five inference phenomena; representations become more classical with task familiarity	Abstract
Broekaert et al. (2017)	E2	Time evolution of mental states in decision-making	Clarifies relations and explanatory value of increasingly complex dynamics	Abstract
Aerts et al. (2017)	E4	Conjunction fallacy	Emergence effects, not order effects, identified as the main mechanism	Abstract
Aerts and Sassoli de Bianchi (2017)	E3	Question order effects and response replicability in sequential survey judgements	Exact fits to both data sets; argues the data are irreducibly non-Hilbertian, that the QQ-equality can hold for non-Hilbertian models, and that order effects plus replicability are jointly modelled	Full text
Busemeyer and Wang (2018)	E3	Context effects across subsets of judged variables	HSM fits compare favourably with Bayesian networks, demonstrating viability	Abstract
Cervantes and Dzhafarov (2018)	E3	Contextuality in paired human choices	Marginal selectivity violated, yet true contextuality detected beyond direct influences	Abstract
Busemeyer et al. (2019)	E3	Belief change during evidence monitoring; sequential probability ratings	Small interference effects (low coherence; 3/11 participants); summed G^2 diff favoured quantum (350, 306, 260, 40), but versus Markov with drift variability quantum wins only at low coherence	Full text
Basieva et al. (2019)	E3	True contextuality versus direct influences in human decisions	Prior claimed demonstrations explained by direct influences; new experiments show true contextuality	Abstract
Busemeyer et al. (2020)	E2	Dynamics of choice, confidence and decision time	Unifies random-walk and quantum-walk dynamics in one process	Abstract
Ozawa and Khrennikov (2020)	E2	Question order effect combined with response replicability	Shows both effects can be jointly modelled, overturning the no-go for this class	Abstract
Yukalov (2020)	E2	Decision making over time; dynamic inconsistency	Argues dynamic inconsistencies are resolved in QDT; quantum-classical correspondence	Abstract
Kvam et al. (2021)	E3	Constructed preference and effect of prior choice on preference strength	Oscillating preference strength and choice-induced shifts captured by the open-system model	Abstract
Ozawa and Khrennikov (2021)	E3	Joint modelling of question order effect, response replicability and QQ-equality	Instruments satisfying QOE, RRE and QQE exist; model closely reproduces Clinton–Gore statistics with an order-independent prior state	Full text
Epping et al. (2023)	E3	Preferential choice and response time	Pure Markov model favoured for most participants; a substantial number required the open system model	Abstract
Khrennikov (2023)	E2	Quantum-like modelling of biological and cognitive information processing	Argues open-system formalism is the natural basis for quantum-like biology and cognition	Abstract
Khrennikov (2024)	E2	Foundational framework for contextual probability update covering classical and quantum measurements	Kolmogorov, von Neumann and quantum-instrument measurements framed within CMM; contextual definitions of order effect, response replicability, interference and Bell-type violations	Full text

Study	Level	Phenomenon	Key result	Basis
Waddup et al. (2023)	E3	Constructive memory; sequential measurement effects	Violation of a temporal Bell inequality observed in one case	Abstract
Velozy and Sobetska (2024)	E3	Conjunction fallacy: coverage of possible probability-judgement patterns in the literature	Data cluster in a few sub-volumes; single fallacy well explored, no-fallacy and double-fallacy regions under-explored	Full text
Huang et al. (2025a)	E3	Probability judgement fallacies	Model compared favourably with the Bayesian Sampler; systematic overestimation of probabilities discovered	Abstract
Huang et al. (2025c)	E2	Recognition heuristic and hindsight bias in paired-comparison judgements	Deterministic heuristic recovered under measurement; probabilistic choice and hindsight bias emerge from superposition and projective update	Full text
Fuyama et al. (2025)	E2	Characterising measurement classes for question order, response replicability and QQ-equality	$SR\bar{P}$ measurements reproduce QOE with RRE and can satisfy or violate the QQ-equality; non-classicality can arise from non-commuting state-update maps with commuting observables	Full text
Asano and Khrennikov (2026)	E2	Temporal dynamics of deliberation and decision, including strategic games	Framework linking GKSL steady states to decisions, stabilisation of non-Nash outcomes, and predicted beat envelopes in conviction	Full text
Bruza et al. (2009)	E2	Word association and cued recall in the mental lexicon	Initial quantum models outlined and experimental tests proposed	Abstract
Blutner et al. (2013)	E2	Borderline vagueness and borderline contradictions	Classical model falls short; quantum interference explains borderline contradictions	Abstract
Pothos et al. (2013)	E2	Similarity judgement (asymmetry, triangle inequality, diagnosticity)	Asymmetry as order effect; triangle-inequality and diagnosticity effects accommodated	Abstract
Trueblood and Hemmer (2017)	E3	Episodic recognition memory: over-distribution and subadditivity	GQEM accounts for over-distribution and subadditivity	Abstract
Busemeyer et al. (2025)	E2	Effect of episodic memory of prior judgements on later judgements (AA, AB/BA, ABA paradigms)	All three accounts cover the combined findings but differ in mechanism (belief-type updates vs non-commuting projections) and in allowing changed answers on repeated questions	Full text
Costello and Watts (2014)	E3	Biases in probability judgement	Near-normative judgements for noise-cancelling expressions; four biases explained by noise	Abstract
Boyer-Kassem et al. (2016)	E3	Conjunction fallacy	Results at odds with quantum-like model predictions	Abstract
Dzhafarov et al. (2016)	E3	Contextuality in behavioural and social data	No evidence of true contextuality in the reviewed datasets	Abstract
Moreira and Wichert (2017)	E3	Question order effects in opinion polls	Both models reproduce the order effects; interference terms play no role, so the classical projection model suffices	Full text
Busemeyer and Wang (2017)	E3	Sequential opinion measurement (ABA design)	Moderate changes in repeated A judgements; correlation pattern consistent with quantum predictions	Abstract
Kellen et al. (2018)	E2	Mirrored question order effects (QQ equality)	Classical models can reproduce mirrored order effects	Abstract
Costello and Watts (2018)	E3	Invariants in probabilistic reasoning (QQ, addition law, Bayes rule identities)	Identities hold despite biases; consistent with noisy classical reasoning	Abstract
Martínez-Martínez and Sánchez-Burillo (2016)	E2	Choice under uncertainty; Prisoner's Dilemma (sure-thing principle violation)	Stationary distribution of the QSW yields choice probabilities that can violate the law of total probability / sure-thing principle; unique stationary state proved (Spohn's theorem)	Abstract
Waddup et al. (2021)	E3	Contextuality in two-person strategic interaction	Evidence of sensitivity to context; quantum model fitted better than a matched classical model	Abstract
Busemeyer et al. (2017)	E2	Neural implementation of quantum cognition operations	Proposes classical neural realisations of quantum-cognition operations (no quantitative results)	Abstract
Khrennikov et al. (2018)	E2	Neural basis of quantum-like decision statistics	Decision state is a steady state reached with exponentially fast convergence (analytical)	Abstract
Khrennikov et al. (2025b)	E2	Neural grounding of quantum-like cognition	Formal coupling proposed; no quantitative results known	Abstract
Khrennikov et al. (2025a)	E2	Neural oscillations as a substrate for quantum-like cognition and AI	Conceptual/formal bridge proposed; no quantitative results known	Abstract
Khrennikov and Yamada (2025)	E2	Neural grounding of quantum-like entanglement in cognition ('mental entanglement')	Formal construction showing how classical neuronal networks can yield QL entangled states and observables; mapping to EEG/MEG functional connectivity measures	Full text

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